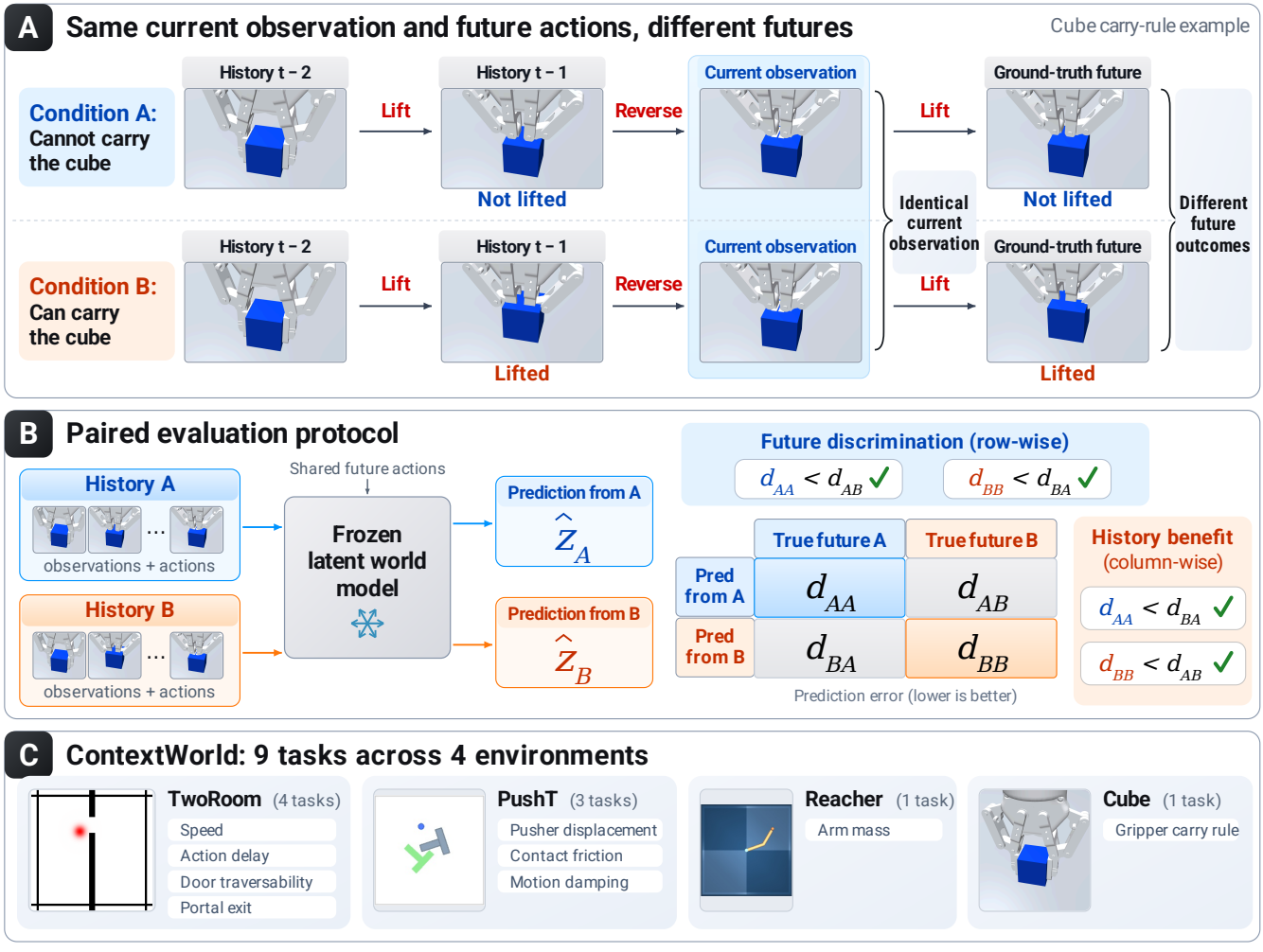

A
Same current observation and future actions, different futures
Cube carry-rule example
Condition A:
Cannot carry the cube
History t − 2
Lift
History t − 1
Not lifted
Reverse
Current observation
Lift
Ground-truth future
Not lifted
Condition B:
Can carry the cube
History t − 2
Lift
History t − 1
Lifted
Reverse
Current observation
Lift
Ground-truth future
Lifted
Identical current observation
Different future outcomes
B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Shared future actions
Frozen latent world model
Prediction from A
$\hat{Z}_A$
Prediction from B
$\hat{Z}_B$
Future discrimination (row-wise)
$d_{AA} < d_{AB}$ ✓
$d_{BB} < d_{BA}$ ✓
True future A
True future B
Pred from A
Pred from B
d_{AA}
d_{AB}
d_{BA}
d_{BB}
Prediction error (lower is better)
History benefit (column-wise)
$d_{AA} < d_{BA}$ ✓
$d_{BB} < d_{AB}$ ✓
C
ContextWorld: 9 tasks across 4 environments
TwoRoom (4 tasks)
Speed
Action delay
Door traversability
Portal exit
PushT (3 tasks)
Pusher displacement
Contact friction
Motion damping
Reacher (1 task)
Arm mass
Cube (1 task)
Gripper carry rule